

Milestone 1

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Python

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Problem Description

This program will allow teachers, or supervisors with some slight modification, to accurately track attendance of students or employees. The intended use would be for administrators, teachers, or supervisors.

The program would allow the user to automatically track absences and attendance with ease. The program would store the inputs inside dictionaries. This would make it easy to expand, allow for loops and conditionals, and have multiple types of variables and lists. This version will be designed for use by a teacher and be able to generate warning messages if a certain number of absences are reached.

Inputs and Outputs

The inputs of the program will ask for the name, date and status of the tracked individual. The outputs would generate a list of names and the number of absences assigned to each name. There can also be an output that generates a warning if any one hits a certain number of absences. This can also be modified to generate different types of messages based on how many absences are reached.

Adding a new input would look like this:

```
record = {  
    "name": "John Smith",  
    "date": "07/11/2026",  
    "status": "Present"  
}
```

`attendance.append(record)` #this adds the record above into the existing list titled attendance.

Creating a variable asking for a person and absences can then be used to print how many absences that person has:

```
person = input("Enter name: ")  
absences = 0
```

for record in attendance:

```
    if record["name"] == person and record["status"] == "absent":
```

```
absences += 1
```

```
print(person + " has " + str(absences) + " absences.")
```

Algorithm Overview

- The program will store multiple values.
- It will gather the input from the user and add it to the correct list.
- It will calculate the number of attendances and absences of a desired individual.
- It will automatically send an alert if certain amounts of absences are reached.

Planned Structure and Functions

The program can have a variety of functions added to it, including counting absences (`count_absences()`), finding all records for one person (`search_member()`), and sending an alert based off absence count (`absence_alert()`).

The `count_absences` would display a number for how many times a searched member has been absent.

The `search_member` would display all pertinent information about the desired member.

The `absence_alert` would generate an alert based on the number of absences an individual has accrued.